SQL Theory and Syntax: Q191 to Q210

Easy interview answers, following the definitions and syntax in your notes

Big picture first (from your notes): SQL has 5 statement types. **DQL** = SELECT. **DDL** = CREATE, RENAME, TRUNCATE, DROP, ALTER (structure, auto-commit). **DML** = INSERT, UPDATE, DELETE (data, non-auto-commit). **TCL** = COMMIT, ROLLBACK, SAVEPOINT. **DCL** = GRANT, REVOKE. Memory: **DDL = Design, DML = Data**.

A. Differences (Q191 to Q193, Q203, Q204)

191) Difference between DELETE and TRUNCATE

Say in interview: DELETE is a DML command that removes specific rows using WHERE. TRUNCATE is a DDL command that removes all records at once; it does not support WHERE and cannot be rolled back.

Point	DELETE	TRUNCATE
Statement type	DML	DDL
WHERE clause	Supported (delete a specific record)	Not supported (all records go)
Rollback	Possible before COMMIT	Not possible (auto-commit)
Speed	Slower, works record by record	Faster
Structure	Stays	Stays
On parent table	Allowed, but child records block it (remove child rows first)	Not possible until the connection to the child table is removed

Easy trick: DELETE = Dustbin with a Decision (WHERE). TRUNCATE = Total wipe, no undo.

192) Difference between DELETE, DROP and TRUNCATE

Say in interview: DELETE removes rows, TRUNCATE removes all rows but keeps the table, DROP removes the whole table from the database and sends it to the recycle bin, so FLASHBACK can bring it back.

Point	DELETE	TRUNCATE	DROP
Removes	Selected or all rows	All rows	Rows and the table itself
Type	DML	DDL	DDL
WHERE	Yes	No	No
Rollback	Yes	No	No, but FLASHBACK can restore it
Table after	Exists	Exists, empty	Gone (in recycle bin)
Permanent removal	-	-	DROP TABLE t PURGE

193) Difference between ALTER and UPDATE

Say in interview: ALTER changes the structure of the table (columns, datatype, constraints). UPDATE changes the values stored in existing records.

ALTER	UPDATE
DDL command, 7 operations (add/drop/rename column, change datatype, NULL/NOT NULL, add/drop constraint)	DML command, changes values of columns in rows
Works on the table structure	Works on the data
Auto-commit	Needs COMMIT
Mostly works on empty tables; few operations work on filled tables	Works on any table, uses WHERE to pick records

203) Difference between Projection and Selection

Say in interview: Projection is fetching data by selecting only the columns. Selection is fetching data by selecting rows (and columns) using a condition.

Projection	Selection
Chooses columns (vertical)	Chooses rows (horizontal)
Uses the SELECT list: *, DISTINCT, column, expression, alias	Uses the WHERE clause
SELECT ename, sal FROM emp;	SELECT * FROM emp WHERE sal > 1000;

204) Difference between DDL and DML

DDL (Data Definition)	DML (Data Manipulation)
Defines or changes structure	Changes the data in the table
CREATE, RENAME, TRUNCATE, DROP, ALTER	INSERT, UPDATE, DELETE
Auto-commit statement (cannot be rolled back)	Non-auto-commit (needs COMMIT, can ROLLBACK)
Affects the whole table	Affects records; WHERE can limit it

B. DDL syntax (Q194, Q197, Q199, Q205 to Q210)

194) Syntax for CREATE TABLE with primary key and foreign key

Say in interview: To create a table we need a table name and all the columns. We validate the schema using datatypes and constraints. Constraints are optional. Create the parent table first, then the child.

```
CREATE TABLE dept (
  deptno NUMBER(2) PRIMARY KEY,                  -- parent table: has the PK
  dname  VARCHAR2(20),
  loc    VARCHAR2(20)
);
CREATE TABLE emp (
  empno  NUMBER(4),
  ename  VARCHAR2(20) NOT NULL,
  sal    NUMBER(8,2) CHECK (sal > 0),
  deptno NUMBER(2),
  CONSTRAINT emp_pk PRIMARY KEY (empno),         -- named PK
  CONSTRAINT emp_fk FOREIGN KEY (deptno) REFERENCES dept(deptno)   -- child table: has the FK
);
```

Table-name rules from your notes: must not start with a number or special character, must be a single string, same column name cannot repeat in a table, same table name cannot repeat in a database.

197) Syntax for all DDL statements

```
CREATE TABLE t1 (id NUMBER(3) PRIMARY KEY, name VARCHAR2(30));
RENAME t1 TO t2;                           -- RENAME old_name TO new_name;
TRUNCATE TABLE t2;                         -- no WHERE, no rollback
DROP TABLE t2;                             -- goes to recycle bin
FLASHBACK TABLE t2 TO BEFORE DROP;         -- get the table back
PURGE TABLE t2;     or     DROP TABLE t2 PURGE;     -- delete even from recycle bin
ALTER TABLE t2 ADD age NUMBER;             -- 7 operations (see Q207 to Q210)
```

Rule to speak: TRUNCATE and DROP are not possible on a parent table until the connection with the child table is removed (using ALTER).

199) Check the table list from the recycle bin

```
SELECT * FROM RECYCLEBIN;      -- tables dropped by you (still recoverable)
SELECT * FROM TAB;             -- tables currently in the database
```

205) Create an index

Say in interview: An index is a structure that speeds up searching on a column, like the index page of a book.

```
CREATE INDEX idx_ename ON emp(ename);
CREATE UNIQUE INDEX idx_empno ON emp(empno);
```

206) Check the structure of a table

```
DESC emp;          -- shows column names, NULL / NOT NULL and datatypes
```

207) Add a column to a table

```
ALTER TABLE emp ADD gender VARCHAR2(1);                -- optional (NULL) column: any table
ALTER TABLE emp ADD mno NUMBER(10) NOT NULL;           -- mandatory column
```

Rule from your notes: adding a mandatory (NOT NULL) column is possible only on an empty table. Adding an optional column is possible for all tables.

208) Drop a column from a table

```
ALTER TABLE emp DROP COLUMN mno;
```

Rule: any column can be dropped, but the primary key of a parent table cannot be dropped until the connection is removed.

209) Change the datatype

```
ALTER TABLE emp MODIFY sal NUMBER(10,2);
ALTER TABLE flight MODIFY mno DATE;
```

Rule: the column must be empty to change the datatype. If it has data, you can only increase the size.

210) Change NULL to NOT NULL or vice versa

```
ALTER TABLE emp MODIFY ename VARCHAR2(30) NOT NULL;
ALTER TABLE emp MODIFY ename VARCHAR2(30) NULL;
```

Rule: NULL to NOT NULL fails if the column already has NULL values. Not applicable to a primary key.

C. DML, TCL, DCL and checking constraints (Q195, Q196, Q198, Q200, Q201, Q202)

196) Syntax for DML

```
INSERT INTO emp VALUES (9001, 'RAVI', 'CLERK', 7902, DATE '2024-01-10', 1200, NULL, 10);
INSERT INTO emp (empno, ename, sal) VALUES (9002, 'ANU', 1300);   -- most commonly used
UPDATE emp SET sal = 5000, job = 'MANAGER' WHERE ename = 'SMITH';
DELETE FROM emp WHERE ename = 'KING';
```

Rule: while inserting into a child table, the FK value must already exist in the parent table. A parent record that has child records cannot be deleted until the connection is removed. DML needs COMMIT.

201) Syntax for TCL statements

Say in interview: TCL controls the transaction in the database. COMMIT saves it, ROLLBACK gets back the transaction before commit, SAVEPOINT controls the transaction in levels.

```
COMMIT;                     -- save the transaction permanently
ROLLBACK;                   -- undo everything before commit
SAVEPOINT sp1;              -- mark a level
ROLLBACK TO sp1;            -- go back to that level only
```

195) Syntax for GRANT and REVOKE (DCL)

Say in interview: DCL controls data flow between multiple users. GRANT gives access to a table, REVOKE takes the access back.

```
GRANT SELECT, INSERT, UPDATE, DELETE ON emp TO scott;       -- GRANT sql_commands ON table TO user;
REVOKE DELETE ON emp FROM scott;                            -- REVOKE sql_command ON table FROM user;
```

From your notes: SELECT access = viewer (read). INSERT / UPDATE / DELETE access = editor (write).

200) Check the permission (access) list (numbered '20' in your list)

```
SELECT * FROM USER_TAB_PRIVS;           -- all access given or received on tables
```

198) Check the constraints

```
SELECT constraint_name, constraint_type, search_condition
FROM user_constraints
WHERE table_name = 'EMP';               -- table name in UPPER CASE
```

Constraint types: **P** = primary key, **R** = foreign key (references), **U** = unique, **C** = check or NOT NULL.

202) Syntax for selection and projection

```
-- Projection: only columns        SELECT * / DISTINCT col / expression [alias] FROM table_name;
SELECT ename, sal*12 AS annual FROM emp;
-- Selection: rows (with condition)
SELECT * FROM emp WHERE deptno = 10;
-- Both together
SELECT ename, sal FROM emp WHERE sal > 1000;
```

Working (from your notes): FROM clause starts the execution and checks the table exists. SELECT then searches the column. If the table is missing: 'table or view does not exist'. If the column is missing: 'invalid identifier'.

Last-minute memory sheet

If asked...	Answer in one line
DELETE vs TRUNCATE vs DROP	Delete = some rows, rollback possible. Truncate = all rows, no rollback. Drop = whole table, recoverable with FLASHBACK until PURGE.
ALTER vs UPDATE	ALTER = structure. UPDATE = data.
DDL vs DML	DDL = structure, auto-commit. DML = data, needs COMMIT.
Projection vs Selection	Projection = columns. Selection = rows.
Get dropped table back	FLASHBACK TABLE t TO BEFORE DROP; Check with SELECT * FROM RECYCLEBIN;
Remove permanently	DROP TABLE t PURGE; or PURGE TABLE t;
TCL	COMMIT, ROLLBACK, SAVEPOINT
DCL	GRANT, REVOKE. Check access: SELECT * FROM USER_TAB_PRIVS;
Table structure	DESC table_name;
Constraint list	SELECT * FROM user_constraints WHERE table_name='EMP';